Develop a customized Ping command to test server connectivity

Aim:

To develop a python program that checks server connectivity using ICMP ping and displays the response time and status.

Requirements:

Hardware:

Computer / Laptop
Internet connection

Software:

Windows / Linux
python 3.10 or above
Vs code / py charm
command Prompt / Terminal

Theory:

Ping is a network diagnostic tool used to check whether a server is reachable. It uses ICMP Echo Request and Reply packets to measure response time and packet loss.

Working Procedure:

Install python and vs code
Create a File named ping.py
Import subprocess and platform
Get the hostname from the user.
Detect the operating system.
Use ping -n 4 on windows and ping -c on Linux/macos

Execute the ping command
Display the ping result
Check the return code
Display Reachable or Unreachable

Program:

```
import subprocess
import platform
print("=" * 40)
print("  Customized Ping Utility")
print("=" * 40)
host = input("Enter Website or IP Address: ")
if platform.system() == "Windows":
    command = ["ping", "-n", "4", host]
else:
    command = ["ping", "-c", "4", host]
try:
    result = subprocess.run(
        command, capture_output=True, text=True
    )
    print("\n Ping Results:\n")
    print(result.stdout)
    if result.returncode == 0:
        print("Server Status: Reachable")
    else:
        print("Server Status: Unreachable")
except Exception as e:
    print("Error:", e)
```

Execution:

open the vs code terminal and run:

python ping.py

Sample Input:

Ping Results:

Reply From google.com:

bytes=32 time=28ms TTL=111

Packets:
Sent = 4
Received = 4
Lost = 0 (0% loss)
Server status: Reachable.

Result:
Thus, the customized ping command was successfully developed in python to test server connectivity and display the ping status.